

Definition and first terms

$31\,A(x) = 30 + 125\,x + A(x)^6$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(5)*T(x)$; $T=x+3*T^2+20*T^3+75*T^4+150*T^5+125*T^6$.
 $a(0),\dots,a(7) = 1, 5, 15, 190, 2550, 38070, 609205, 10199640$.
 Kernel used throughout: $\rho(u) = u\,(1-3\,u^1-20\,u^2-75\,u^3-150\,u^4-125\,u^5)$, so $x = \rho(T(x))$.

Typogeometry

1x5

{1,1}x15

{1,{1,1}}x45

{1,1,1}x100

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 3, Δ_3 : 20, Δ_4 : 75, Δ_5 : 150, Δ_6 : 125.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1,\dots,k_5) \in \mathbb{Z}_{\geq 0}^5 : \sum_{j=1}^5 j k_j = n-1\}, \quad K = \sum_{j=1}^5 k_j,$$

$$a(n) = \frac{5}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n+K-1}{K} \binom{K}{k_1,\dots,k_5} 3^{k_1} 20^{k_2} 75^{k_3} 150^{k_4} 125^{k_5}.$$

$$a(n) = \frac{5}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^5 P_r(n)a(n+r) = 0$ for $n \geq 1$. The exact degree-5 coefficient polynomials are embedded in `certificate_payload.json` (source: `data/recurrence.json`).

Differential equation

The scalar linear ODE has order 5. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `data/ode.json`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n,u) = N(n,u)/\rho(u)^4$, $\deg_u N = 25$.
 Telescoping identity: $\sum_r P_r(n)H_{n+r}(u) = \partial_u(R(n,u)H_n(u))$.
 Matrix dimensions: 12×12 ; remainder 5×6 , rank 5, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.